

Jaelan Cruz

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SUMMARY

Full-Stack Engineer and Computer Science student focused on building reliable **high-performance interactive** systems with emphasis on **scalability** and **maintainability**.

EDUCATION

York University

Bachelor of Applied Science – CSSD

- **Relevant Coursework:** Data Structures & Algorithms, User Interfaces, Software Design, Object-Oriented Programming, Intro to Cloud.

Markham, ON

Expected Apr. 2028

EXPERIENCE

Fullstack Engineer

Dec 2025 – Present

Circe – Contract Part-Time

- Collaborated with **6+ contributors** on a production **React Native app**, resolving merge conflicts and maintaining code quality.
- Built and hardened a **REST API** with **FastAPI** and **PostgreSQL** for secure, scalable user profile management, resulting in reliable authentication and **robust data protection**.
- Coupled client-side software to internal systems via **TypeScript API client** and custom **React hooks**, handling loading, error, and update states, which improved real-time data accuracy and **user experience**.
- Engineered **5 authentication flows** (sign-in, sign-up, confirmation, password reset, forgot password) with robust validation and real-time error messaging, enhancing **account security** and reducing **user errors**.
- Refactored **2,000+ lines** across **8+ files** for **reduced duplication** and **streamlined development**, which simplified maintenance and enabled faster feature updates.

PROJECTS

Interactive 3D Portfolio Experience | *Astro, Svelte, Three.js, GSAP, TailwindCSS*

- Engineered a **modular, checkpoint-driven** 3D portfolio with reusable scene systems; galaxy renders **25,000+ particles** and a **1,800-star** field using a single merged **Points** buffer.
- Built **GSAP** timelines and an **AudioManager** to synchronize animations and SFX across **six checkpoint** sections for smooth, **scroll-driven** guided interactions.
- Reduced runtime cost via **chunked buffer updates** (**12%** of particles per frame, min **1.2k**), **renderer tuning**, and **depth-based** size/opacity adjustments.
- Automated builds and deployments using **GitHub Actions** (**build** → **artifact** → **deploy**) for repeatable production releases.

jamSpace | *Python, Flask, HTML, CSS, JavaScript*

- Developed a full-stack productivity web application using **Flask** and modular **Blueprint architecture** for **scalable route management**.
- Removed **5+ obsolete files** and reduced codebase clutter by **1,000+ lines** via systematic cleanup.
- Enhanced **UI/UX** with custom CSS/JavaScript for **responsive design**, modals, and advanced animations.

PixelCraft | *Java, Bash, OOP, CLI, HTML, CSS*

- Developed an image processing tool supporting batch transformations with **8+ filters**, processing **1,000+ images** in testing.
- Designed extensible **OOP** architecture using **Java Reflection** for dynamic filter selection and rapid addition of new effects.
- Implemented a **command-line interface** and **10+ Bash scripts** for flexible usage and **100% automated batch workflows**.
- Processed **PNG images** and automated file management for efficient, large-scale operations.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, SQL, JavaScript, TypeScript, HTML5, CSS, Bash

Libraries & Frameworks: React, FastAPI, Astro, Svelte, Three.js, GSAP, WebGL, Flask, TailwindCSS, Formik, Yup, Jinja2

Databases: PostgreSQL, SQLite

Tools & Platforms: Git, GitHub, AWS, Docker, VS Code, Linux, GitHub Actions, Jest, pytest, Maven, Expo, CMake

AI & Automation: GPT-5 mini, Claude Sonnet 4.6, GPT-5.3 Codex, Copilot